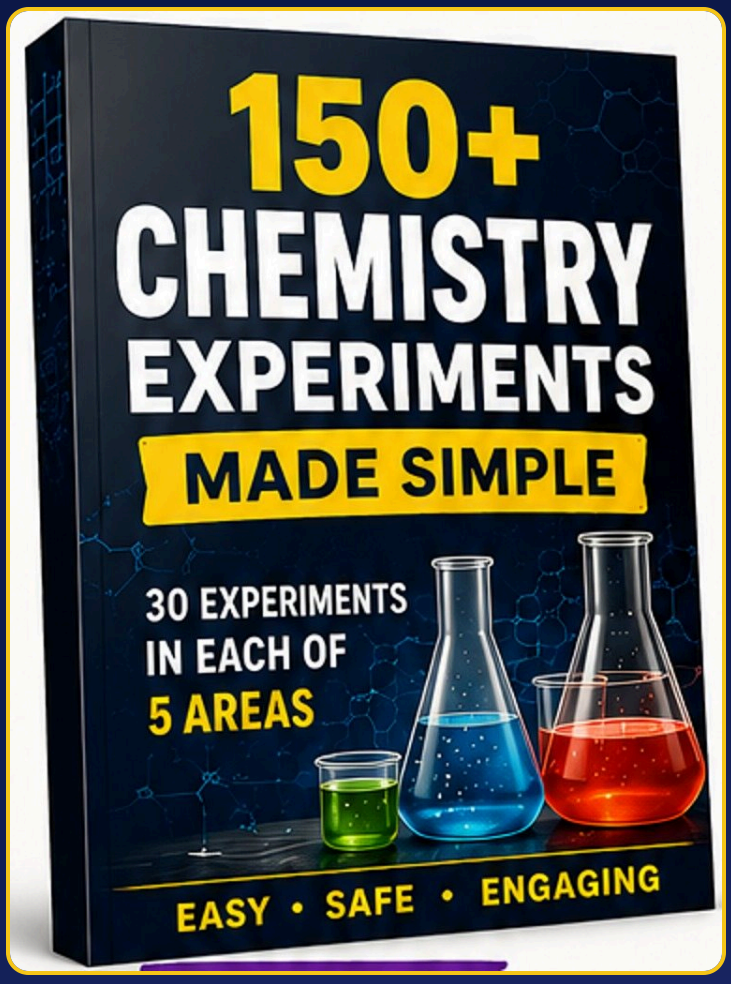
BONUS 3

Visual Periodic Table + 12 Classroom Infographics

A poster-size periodic table with everyday uses, plus 12 posters for your walls



The Visual Periodic Table

118 elements with the everyday uses you already know

1 **H** **Hydrogen** 1.008 Balloons and rocket fuel																	2 **He** **Helium** 4.003 Party balloons and MRI cooling
3 **Li** **Lithium** 6.94 Phone and laptop batteries	4 **Be** **Beryllium** 9.012 Aircraft parts and X-ray windows											5 **B** **Boron** 10.81 Laundry booster and heat-proof glass	6 **C** **Carbon** 12.011 Pencils, diamonds and all living things	7 **N** **Nitrogen** 14.007 Fertilizer and food packaging	8 **O** **Oxygen** 15.999 Breathing and hospital oxygen tanks	9 **F** **Fluorine** 18.998 Toothpaste and nonstick pans	10 **Ne** **Neon** 20.180 Glowing signs in shop windows
11 **Na** **Sodium** 22.990 Table salt and nerve signals	12 **Mg** **Magnesium** 24.305 Sparklers and antacid tablets											13 **Al** **Aluminum** 26.982 Soda cans and kitchen foil	14 **Si** **Silicon** 28.085 Computer chips and glass	15 **P** **Phosphorus** 30.974 Matches and plant fertilizer	16 **S** **Sulfur** 32.06 Matches and car tires	17 **Cl** **Chlorine** 35.45 Swimming pool water and bleach	18 **Ar** **Argon** 39.948 Light bulbs and welding
19 **K** **Potassium** 39.098 Bananas and plant fertilizer	20 **Ca** **Calcium** 40.078 Bones, teeth and chalk	21 **Sc** **Scandium** 44.956 Stadium lights and bike frames	22 **Ti** **Titanium** 47.867 Bike frames and joint implants	23 **V** **Vanadium** 50.942 Strong steel tools	24 **Cr** **Chromium** 51.996 Shiny chrome car trim	25 **Mn** **Manganese** 54.938 Steel and batteries	26 **Fe** **Iron** 55.845 Steel bridges, cars and tools	27 **Co** **Cobalt** 58.933 Blue pottery and phone batteries	28 **Ni** **Nickel** 58.693 Coins and stainless steel	29 **Cu** **Copper** 63.546 Electric wires and water pipes	30 **Zn** **Zinc** 65.38 Rust-proof coatings and sunscreen	31 **Ga** **Gallium** 69.723 Phone chips and LED lights	32 **Ge** **Germanium** 72.630 Fiber-optic cables and camera lenses	33 **As** **Arsenic** 74.922 Computer chips and old wood treatments	34 **Se** **Selenium** 78.971 Dandruff shampoo and colored glass	35 **Br** **Bromine** 79.904 Fire retardants and pool cleaners	36 **Kr** **Krypton** 83.798 Bright photo flash lamps
37 **Rb** **Rubidium** 85.468 Atomic clocks and research	38 **Sr** **Strontium** 87.62 Red fireworks and road flares	39 **Y** **Yttrium** 88.906 LED lights and old TV screens	40 **Zr** **Zirconium** 91.224 Fake diamonds and nuclear fuel tubes	41 **Nb** **Niobium** 92.906 MRI magnets and jet engines	42 **Mo** **Molybdenum** 95.95 Strong steel for tools and engines	43 **Tc** **Technetium** [98] Medical scans as a radioactive tracer	44 **Ru** **Ruthenium** 101.07 Pen tips and electronics	45 **Rh** **Rhodium** 102.91 Car exhaust cleaners	46 **Pd** **Palladium** 106.42 Car exhaust cleaners and jewelry	47 **Ag** **Silver** 107.87 Jewelry, mirrors and coins	48 **Cd** **Cadmium** 112.41 Yellow paint and rechargeable batteries	49 **In** **Indium** 114.82 Touchscreens and flat TVs	50 **Sn** **Tin** 118.71 Food cans and solder	51 **Sb** **Antimony** 121.76 Flame-proofing and batteries	52 **Te** **Tellurium** 127.60 Solar panels and rewritable discs	53 **I** **Iodine** 126.90 Table salt additive and antiseptic	54 **Xe** **Xenon** 131.29 Car headlights and camera flashes
55 **Cs** **Cesium** 132.91 Atomic clocks	56 **Ba** **Barium** 137.33 X-ray meals and green fireworks	**57-71** **Lanthanides**	72 **Hf** **Hafnium** 178.49 Computer chips and nuclear control rods	73 **Ta** **Tantalum** 180.95 Phone and laptop capacitors	74 **W** **Tungsten** 183.84 Old light bulb filaments	75 **Re** **Rhenium** 186.21 Jet engine parts	76 **Os** **Osmium** 190.23 Fountain pen tips	77 **Ir** **Iridium** 192.22 Spark plugs and pen nibs	78 **Pt** **Platinum** 195.08 Jewelry and car exhaust cleaners	79 **Au** **Gold** 196.97 Jewelry, coins and phone contacts	80 **Hg** **Mercury** 200.59 Old thermometers and some lamps	81 **Tl** **Thallium** 204.38 Research only	82 **Pb** **Lead** 207.2 Car batteries and radiation shields	83 **Bi** **Bismuth** 208.98 Stomach medicine and cosmetics	84 **Po** **Polonium** [209] Static-removing brushes (rare)	85 **At** **Astatine** [210] Research only	86 **Rn** **Radon** [222] Home radon test kits
87 **Fr** **Francium** [223] Research only	88 **Ra** **Radium** [226] Old glow-in-the-dark watch dials	**89-103** **Actinides**	104 **Rf** **Rutherfordium** [267] Made in labs; not found in daily life	105 **Db** **Dubnium** [268] Made in labs; not found in daily life	106 **Sg** **Seaborgium** [269] Made in labs; not found in daily life	107 **Bh** **Bohrium** [270] Made in labs; not found in daily life	108 **Hs** **Hassium** [277] Made in labs; not found in daily life	109 **Mt** **Meitnerium** [278] Made in labs; not found in daily life	110 **Ds** **Darmstadtium** [281] Made in labs; not found in daily life	111 **Rg** **Roentgenium** [282] Made in labs; not found in daily life	112 **Cn** **Copernicium** [285] Made in labs; not found in daily life	113 **Nh** **Nihonium** [286] Made in labs; not found in daily life	114 **Fl** **Flerovium** [289] Made in labs; not found in daily life	115 **Mc** **Moscovium** [290] Made in labs; not found in daily life	116 **Lv** **Livermorium** [293] Made in labs; not found in daily life	117 **Ts** **Tennessine** [294] Made in labs; not found in daily life	118 **Og** **Oganesson** [294] Made in labs; not found in daily life
		57 **La** **Lanthanum** 138.91 Camera lenses and lighter flints	58 **Ce** **Cerium** 140.12 Lighter flints and glass polish	59 **Pr** **Praseodymium** 140.91 Yellow glass and strong magnets	60 **Nd** **Neodymium** 144.24 Strong magnets in headphones	61 **Pm** **Promethium** [145] Old glow-in-the-dark dials	62 **Sm** **Samarium** 150.36 Strong magnets and headphones	63 **Eu** **Europium** 151.96 Euro banknote security ink	64 **Gd** **Gadolinium** 157.25 MRI scan contrast liquid	65 **Tb** **Terbium** 158.93 Green color in screens and lights	66 **Dy** **Dysprosium** 162.50 Magnets in electric cars	67 **Ho** **Holmium** 164.93 Lasers and strong magnets	68 **Er** **Erbium** 167.26 Fiber-optic signal boosters	69 **Tm** **Thulium** 168.93 Portable X-ray machines	70 **Yb** **Ytterbium** 173.05 Atomic clocks and steel	71 **Lu** **Lutetium** 174.97 Medical PET scan detectors	
		89 **Ac** **Actinium** [227] Research only	90 **Th** **Thorium** 232.04 Old gas lantern mantles	91 **Pa** **Protactinium** 231.04 Research only	92 **U** **Uranium** 238.03 Nuclear power plants	93 **Np** **Neptunium** [237] Research only	94 **Pu** **Plutonium** [244] Power for space probes	95 **Am** **Americium** [243] Smoke detectors	96 **Cm** **Curium** [247] Research only	97 **Bk** **Berkelium** [247] Research only	98 **Cf** **Californium** [251] Research only	99 **Es** **Einsteinium** [252] Research only	100 **Fm** **Fermium** [257] Research only	101 **Md** **Mendelevium** [258] Research only	102 **No** **Nobelium** [259] Research only	103 **Lr** **Lawrencium** [266] Research only	

Alkali Metal · Alkaline Earth Metal · Transition Metal · Post-Transition Metal · Metalloid · Nonmetal · Halogen · Noble Gas · Lanthanide · Actinide

Using this bonus

This bonus has two parts: a **poster-size periodic table** (36 in x 24 in, print it at a copy shop or print it on letter paper to read) with an everyday use for every element, and **12 classroom infographics** (Letter size) for your walls.

CLASSROOM INFOGRAPHIC 1 OF 12

States of Matter

Solid, liquid, gas and beyond



Solid

Particles are packed tightly and only vibrate. A solid keeps its own shape.



Liquid

Particles slide past each other. A liquid takes the shape of its container.



Gas

Particles move fast and spread out. A gas fills any space it is given.



Melting and Freezing

Heat turns a solid into a liquid. Removing heat turns it back.



Evaporation and Condensation

A liquid becomes a gas when heated, and a gas becomes a liquid when cooled.



Plasma

The fourth state of matter is a hot, glowing gas of charged particles, found in lightning and the Sun.

Matter changes state when heat is added or removed.

CLASSROOM INFOGRAPHIC 2 OF 12

Atoms and Molecules

The tiny building blocks of everything

The Atom

An atom is the smallest unit of an element that still has that element's properties.

The Nucleus

Protons and neutrons are packed together in the center of the atom.

Electrons

Tiny electrons whiz around the nucleus and decide how atoms bond.

Elements

An element contains only one kind of atom, like oxygen or gold.

Molecules

A molecule is two or more atoms joined together, like water, H2O.

Compounds

A compound is a molecule made of different elements, like salt or sugar.

Everything around you is built from atoms.

CLASSROOM INFOGRAPHIC 3 OF 12

Physical vs Chemical Changes

Same substance or a brand new one?

Physical Change

A physical change alters form or appearance, but not what the substance is.

Physical Examples

Melting ice, tearing paper and dissolving sugar are physical changes.

Chemical Change

A chemical change makes one or more new substances with new properties.

Chemical Examples

Burning wood, rusting iron and baking a cake are chemical changes.

Can You Undo It?

Physical changes are often easy to reverse. Chemical changes usually are not.

Look for Clues

Color change, bubbles, light or a new smell hint at a chemical change.

Chemical changes make new substances; physical changes do not.

CLASSROOM INFOGRAPHIC 4 OF 12

The pH Scale

How acidic or basic is it?

The Scale

The pH scale runs from 0 to 14. Below 7 is acidic and above 7 is basic.

Neutral at 7

Pure water sits at about pH 7, right in the middle.

Acidic Foods

Lemon juice is around pH 2 and vinegar is around pH 2.5 to 3.

Almost Neutral

Milk is slightly acidic at about pH 6.5, close to neutral.

Basic Helpers

Baking soda is about pH 9, and hand soap is often 9 to 10.

Strong Cleaners

Household bleach is around pH 12 to 13, so handle it with care.

Each step on the pH scale is a ten times difference.

CLASSROOM INFOGRAPHIC 5 OF 12

Signs of a Chemical Reaction

How to tell something new has formed



Color Change

A new color can mean new substances have formed.



Gas Bubbles

Fizzing or bubbling without heating often means a gas is being made.



Solid Forms

A solid that appears in a clear liquid is called a precipitate.



Temperature Change

The mixture gets hotter or colder on its own.



Light or Flame

Some reactions give off light, like a glow stick or a fire.



New Smell

A new odor can show that a new substance has been made.

Look for several signs together to be sure.

CLASSROOM INFOGRAPHIC 6 OF 12

Safety Symbols and Lab Rules

Stay safe while you explore

Wear Goggles

Safety goggles protect your eyes from splashes. Wear them every time.

Flammable Symbol

A flame symbol means the material can catch fire. Keep it away from heat.

Corrosive Symbol

This symbol warns that a substance can burn skin or damage materials.

Toxic Symbol

A skull symbol means a substance is poisonous. Never taste or sniff it.

No Eating or Tasting

Never eat, drink or taste anything in the lab.

Tell Your Teacher

Report spills, breaks and injuries right away, and wash your hands at the end.

When in doubt, stop and ask.

CLASSROOM INFOGRAPHIC 7 OF 12

Lab Tools and Their Uses

The equipment every chemist knows



Beaker

A beaker holds and mixes liquids. Its markings are only rough estimates.



Graduated Cylinder

This tall tube measures liquid volume accurately.



Test Tube

Small test tubes hold tiny amounts for reactions.



Balance

A balance measures mass, how much matter something has.



Thermometer

A thermometer measures temperature.



Dropper

A dropper moves small amounts of liquid one drop at a time.

Choose the right tool for each measurement.

CLASSROOM INFOGRAPHIC 8 OF 12

The Scientific Method

A step-by-step way to find answers



Ask a Question

Start with something you notice and want to understand.



Do Some Research

Find out what is already known about the topic.



Make a Hypothesis

Write a testable prediction that can be proved right or wrong.



Run an Experiment

Change one thing at a time and keep everything else the same.



Record and Analyze

Collect data carefully and look for patterns in your results.



Share Conclusions

Explain what you learned, even when the hypothesis was wrong.

Good science tests ideas and follows the evidence.

CLASSROOM INFOGRAPHIC 9 OF 12

Mixtures and Solutions

Blending substances without bonding



What Is a Mixture

A mixture combines substances that keep their own properties.



Heterogeneous

You can see the different parts, like in a salad or trail mix.



Homogeneous

It looks the same throughout, like air or salt water.



Solute and Solvent

In a solution, the solute dissolves in the solvent. In salt water, salt is the solute.



Separating Mixtures

Filtering, evaporating and using magnets can separate mixtures.



Suspensions

Particles in a suspension, like muddy water, settle out if left alone.

In a mixture, no new substance is made.

CLASSROOM INFOGRAPHIC 10 OF 12

Metals, Nonmetals and Metalloids

Three families of elements

Metals

Most elements are metals. They are shiny, bendable and good conductors.

Nonmetals

Nonmetals are usually dull and brittle as solids, and poor conductors.

Metalloids

Metalloids such as silicon sit in between and mix both traits.

Where They Sit

On the periodic table, metals are on the left and nonmetals on the right.

Everyday Metals

Iron in bridges, copper in wires and aluminum in cans are metals.

Silicon Chips

Silicon is a metalloid and is the key material in computer chips.

Properties help us sort the elements into families.

CLASSROOM INFOGRAPHIC 11 OF 12

Acids and Bases

Opposites that cancel each other out



Acids

Acids taste sour, like lemon juice. They have a pH below 7.



Bases

Bases feel slippery and taste bitter, like soap. They have a pH above 7.



Indicators

Indicators such as litmus paper or red cabbage juice change color to show pH.



Neutralization

An acid and a base react to make water and a salt.



Everyday Examples

Vinegar is an acid, and baking soda is a base.



Never Taste

Never taste or touch lab chemicals to test them.

Acids and bases react to balance each other.

CLASSROOM INFOGRAPHIC 12 OF 12

Conservation of Mass

Matter is never lost

The Law

In a chemical reaction, mass is neither created nor destroyed.

Lavoisier

Antoine Lavoisier measured reactions carefully in the 1700s to show this.

Atoms Rearrange

Atoms in the reactants are only regrouped into new products.

Burning a Candle

A candle seems to vanish, but its mass leaves as gases like carbon dioxide and water vapor.

Closed Systems

If nothing can escape, the total mass before and after is the same.

Balanced Equations

Chemists balance equations so each side has the same atoms.

Total mass before a reaction equals total mass after.

Element details

Z	SYM	NAME	ATOMIC MASS	CATEGORY	EVERYDAY USE	FUN FACT
1	H	Hydrogen	1.008	nonmetal	Balloons and rocket fuel	The most abundant element in the universe.
2	He	Helium	4.003	noble gas	Party balloons and MRI cooling	Found on the Sun in 1868, years before it was found on Earth.
3	Li	Lithium	6.94	alkali metal	Phone and laptop batteries	The least dense metal; it is light enough to float on oil.
4	Be	Beryllium	9.012	alkaline earth metal	Aircraft parts and X-ray windows	Emeralds and aquamarines are gems that contain beryllium.
5	B	Boron	10.81	metalloid	Laundry booster and heat-proof glass	Boron compounds give fireworks a green flame.
6	C	Carbon	12.011	nonmetal	Pencils, diamonds and all living things	Diamond and pencil graphite are both pure carbon.
7	N	Nitrogen	14.007	nonmetal	Fertilizer and food packaging	It makes up about 78% of the air we breathe.
8	O	Oxygen	15.999	nonmetal	Breathing and hospital oxygen tanks	It is the most abundant element in Earth's crust by mass.
9	F	Fluorine	18.998	halogen	Toothpaste and nonstick pans	It is the most reactive of all the elements.
10	Ne	Neon	20.180	noble gas	Glowing signs in shop windows	Neon glows reddish-orange when electricity passes through it.
11	Na	Sodium	22.990	alkali metal	Table salt and nerve signals	A piece of sodium reacts fiercely when dropped in water.
12	Mg	Magnesium	24.305	alkaline earth metal	Sparklers and antacid tablets	Burning magnesium makes a brilliant white light.
13	Al	Aluminum	26.982	post-transition metal	Soda cans and kitchen foil	It is the most abundant metal in Earth's crust.
14	Si	Silicon	28.085	metalloid	Computer chips and glass	After oxygen, it is the second most abundant element in Earth's crust.
15	P	Phosphorus	30.974	nonmetal	Matches and plant fertilizer	It was the first element discovered by a person, in 1669, from urine.
16	S	Sulfur	32.06	nonmetal	Matches and car tires	Burning sulfur gives off an eerie blue flame.
17	Cl	Chlorine	35.45	halogen	Swimming pool water and bleach	A poisonous gas on its own, yet half of table salt.
18	Ar	Argon	39.948	noble gas	Light bulbs and welding	It is the third most common gas in air, about 0.93%.
19	K	Potassium	39.098	alkali metal	Bananas and plant fertilizer	Bananas are very slightly radioactive because of potassium-40.
20	Ca	Calcium	40.078	alkaline earth metal	Bones, teeth and chalk	It is the most abundant metal in the human body.
21	Sc	Scandium	44.956	transition metal	Stadium lights and bike frames	Mendeleev predicted it years before it was discovered.
22	Ti	Titanium	47.867	transition metal	Bike frames and joint implants	It is as strong as steel but much lighter.
23	V	Vanadium	50.942	transition metal	Strong steel tools	Named after Vanadis, a Scandinavian goddess.

Z	SYM	NAME	ATOMIC MASS	CATEGORY	EVERYDAY USE	FUN FACT
24	Cr	Chromium	51.996	transition metal	Shiny chrome car trim	Chromium gives rubies their red color.
25	Mn	Manganese	54.938	transition metal	Steel and batteries	Your body needs a tiny amount to stay healthy.
26	Fe	Iron	55.845	transition metal	Steel bridges, cars and tools	Iron in your blood helps carry oxygen.
27	Co	Cobalt	58.933	transition metal	Blue pottery and phone batteries	It is at the center of vitamin B12.
28	Ni	Nickel	58.693	transition metal	Coins and stainless steel	The US five-cent coin is 75% copper and only 25% nickel.
29	Cu	Copper	63.546	transition metal	Electric wires and water pipes	The Statue of Liberty is green because copper slowly weathers.
30	Zn	Zinc	65.38	transition metal	Rust-proof coatings and sunscreen	Brass is made from copper and zinc.
31	Ga	Gallium	69.723	post-transition metal	Phone chips and LED lights	It melts in your hand, at about 30 degrees Celsius.
32	Ge	Germanium	72.630	metalloid	Fiber-optic cables and camera lenses	Mendeleev predicted it and called it ekasilicon.
33	As	Arsenic	74.922	metalloid	Computer chips and old wood treatments	It is famous as a poison in mystery stories.
34	Se	Selenium	78.971	nonmetal	Dandruff shampoo and colored glass	Named after Selene, the Greek goddess of the Moon.
35	Br	Bromine	79.904	halogen	Fire retardants and pool cleaners	It is one of only two elements that are liquid at room temperature.
36	Kr	Krypton	83.798	noble gas	Bright photo flash lamps	Its name means hidden in Greek.
37	Rb	Rubidium	85.468	alkali metal	Atomic clocks and research	Named from the Latin for deep red, after its spectral lines.
38	Sr	Strontium	87.62	alkaline earth metal	Red fireworks and road flares	Strontium makes fireworks burn crimson red.
39	Y	Yttrium	88.906	transition metal	LED lights and old TV screens	Named after Ytterby, a village in Sweden.
40	Zr	Zirconium	91.224	transition metal	Fake diamonds and nuclear fuel tubes	Zircon crystals are among the oldest minerals on Earth.
41	Nb	Niobium	92.906	transition metal	MRI magnets and jet engines	Named after Niobe, a figure from Greek myth.
42	Mo	Molybdenum	95.95	transition metal	Strong steel for tools and engines	Its name comes from the Greek word for lead.
43	Tc	Technetium	[98]	transition metal	Medical scans as a radioactive tracer	It was the first element made artificially.
44	Ru	Ruthenium	101.07	transition metal	Pen tips and electronics	Named after Ruthenia, an old name for Russia.
45	Rh	Rhodium	102.91	transition metal	Car exhaust cleaners	It is one of the most expensive metals in the world.
46	Pd	Palladium	106.42	transition metal	Car exhaust cleaners and jewelry	It can soak up hundreds of times its volume in hydrogen.
47	Ag	Silver	107.87	transition metal	Jewelry, mirrors and coins	It is the best electrical conductor of all metals.
48	Cd	Cadmium	112.41	transition metal	Yellow paint and rechargeable batteries	It is toxic, so many uses have been phased out.
49	In	Indium	114.82	post-transition metal	Touchscreens and flat TVs	Named for the indigo-blue line in its spectrum.

Z	SYM	NAME	ATOMIC MASS	CATEGORY	EVERYDAY USE	FUN FACT
50	Sn	Tin	118.71	post-transition metal	Food cans and solder	It has 10 stable isotopes, more than any other element.
51	Sb	Antimony	121.76	metalloid	Flame-proofing and batteries	Ancient Egyptians used it in eye makeup.
52	Te	Tellurium	127.60	metalloid	Solar panels and rewritable discs	Named after Tellus, the Latin name for Earth.
53	I	Iodine	126.90	halogen	Table salt additive and antiseptic	Your thyroid needs iodine to work properly.
54	Xe	Xenon	131.29	noble gas	Car headlights and camera flashes	It is used in ion engines on some spacecraft.
55	Cs	Cesium	132.91	alkali metal	Atomic clocks	The second is defined by a cesium atom's vibrations.
56	Ba	Barium	137.33	alkaline earth metal	X-ray meals and green fireworks	Barium makes fireworks burn green.
57	La	Lanthanum	138.91	lanthanide	Camera lenses and lighter flints	Its name means to lie hidden in Greek.
58	Ce	Cerium	140.12	lanthanide	Lighter flints and glass polish	It is the most abundant of the rare earth elements.
59	Pr	Praseodymium	140.91	lanthanide	Yellow glass and strong magnets	Its name means green twin in Greek.
60	Nd	Neodymium	144.24	lanthanide	Strong magnets in headphones	Its magnets are among the strongest permanent magnets.
61	Pm	Promethium	[145]	lanthanide	Old glow-in-the-dark dials	Named after Prometheus, who stole fire in Greek myth.
62	Sm	Samarium	150.36	lanthanide	Strong magnets and headphones	It was named after a mineral, samarskite, named for a mining official.
63	Eu	Europium	151.96	lanthanide	Euro banknote security ink	It was named after the continent of Europe.
64	Gd	Gadolinium	157.25	lanthanide	MRI scan contrast liquid	Named after the chemist Johan Gadolin.
65	Tb	Terbium	158.93	lanthanide	Green color in screens and lights	Named after Ytterby, a village in Sweden.
66	Dy	Dysprosium	162.50	lanthanide	Magnets in electric cars	Its name means hard to get in Greek.
67	Ho	Holmium	164.93	lanthanide	Lasers and strong magnets	Named after Stockholm, whose Latin name is Holmia.
68	Er	Erbium	167.26	lanthanide	Fiber-optic signal boosters	Named after Ytterby, a village in Sweden.
69	Tm	Thulium	168.93	lanthanide	Portable X-ray machines	Named after Thule, an ancient name for Scandinavia.
70	Yb	Ytterbium	173.05	lanthanide	Atomic clocks and steel	Named after Ytterby, a village in Sweden.
71	Lu	Lutetium	174.97	lanthanide	Medical PET scan detectors	Named after Lutetia, the Roman name for Paris.
72	Hf	Hafnium	178.49	transition metal	Computer chips and nuclear control rods	Named after Hafnia, the Latin name for Copenhagen.
73	Ta	Tantalum	180.95	transition metal	Phone and laptop capacitors	Named after Tantalus from Greek myth.
74	W	Tungsten	183.84	transition metal	Old light bulb filaments	It has the highest melting point of any metal, 3422 degrees Celsius.
75	Re	Rhenium	186.21	transition metal	Jet engine parts	It was the last stable element to be discovered, in 1925.

Z	SYM	NAME	ATOMIC MASS	CATEGORY	EVERYDAY USE	FUN FACT
76	Os	Osmium	190.23	transition metal	Fountain pen tips	It is the densest naturally occurring element.
77	Ir	Iridium	192.22	transition metal	Spark plugs and pen nibs	A layer of iridium in rocks marks the asteroid that ended the dinosaurs.
78	Pt	Platinum	195.08	transition metal	Jewelry and car exhaust cleaners	It is used in cisplatin, a cancer-fighting drug.
79	Au	Gold	196.97	transition metal	Jewelry, coins and phone contacts	Gold never rusts or tarnishes.
80	Hg	Mercury	200.59	transition metal	Old thermometers and some lamps	It is the only metal that is liquid at room temperature.
81	Tl	Thallium	204.38	post-transition metal	Research only	Its name comes from the Greek for green twig, after a green spectral line.
82	Pb	Lead	207.2	post-transition metal	Car batteries and radiation shields	Its symbol Pb comes from the Latin word plumbum.
83	Bi	Bismuth	208.98	post-transition metal	Stomach medicine and cosmetics	Bismuth crystals grow in rainbow-colored stair-step shapes.
84	Po	Polonium	[209]	post-transition metal	Static-removing brushes (rare)	Marie Curie named it after her homeland, Poland.
85	At	Astatine	[210]	halogen	Research only	It is the rarest naturally occurring element in Earth's crust.
86	Rn	Radon	[222]	noble gas	Home radon test kits	This radioactive gas seeps up from rocks and soil.
87	Fr	Francium	[223]	alkali metal	Research only	It is extremely rare and only lasts minutes before decaying.
88	Ra	Radium	[226]	alkaline earth metal	Old glow-in-the-dark watch dials	It was discovered by Marie and Pierre Curie.
89	Ac	Actinium	[227]	actinide	Research only	It glows blue in the dark because of its radioactivity.
90	Th	Thorium	232.04	actinide	Old gas lantern mantles	Named after Thor, the Norse god of thunder.
91	Pa	Protactinium	231.04	actinide	Research only	Its name means parent of actinium.
92	U	Uranium	238.03	actinide	Nuclear power plants	One small uranium fuel pellet holds as much energy as about a ton of coal.
93	Np	Neptunium	[237]	actinide	Research only	It was the first element found beyond uranium, named after Neptune.
94	Pu	Plutonium	[244]	actinide	Power for space probes	Plutonium-238 powers Voyager and Mars rovers.
95	Am	Americium	[243]	actinide	Smoke detectors	A tiny amount of it sits inside many smoke alarms.
96	Cm	Curium	[247]	actinide	Research only	Named in honor of Marie and Pierre Curie.
97	Bk	Berkelium	[247]	actinide	Research only	Named after Berkeley, California.
98	Cf	Californium	[251]	actinide	Research only	Named after California.
99	Es	Einsteinium	[252]	actinide	Research only	It was found in the debris of the first hydrogen bomb test.
100	Fm	Fermium	[257]	actinide	Research only	Named after the physicist Enrico Fermi.
101	Md	Mendelevium	[258]	actinide	Research only	Named after Dmitri Mendeleev, who created the periodic table.

Z	SYM	NAME	ATOMIC MASS	CATEGORY	EVERYDAY USE	FUN FACT
102	No	Nobelium	[259]	actinide	Research only	Named after Alfred Nobel, founder of the Nobel Prizes.
103	Lr	Lawrencium	[266]	actinide	Research only	Named after Ernest Lawrence, inventor of the cyclotron.
104	Rf	Rutherfordium	[267]	transition metal	Made in labs; not found in daily life	Named after physicist Ernest Rutherford.
105	Db	Dubnium	[268]	transition metal	Made in labs; not found in daily life	Named after Dubna, the Russian town where it was studied.
106	Sg	Seaborgium	[269]	transition metal	Made in labs; not found in daily life	Named after Glenn Seaborg while he was still alive.
107	Bh	Bohrium	[270]	transition metal	Made in labs; not found in daily life	Named after physicist Niels Bohr.
108	Hs	Hassium	[277]	transition metal	Made in labs; not found in daily life	Named after Hesse, a state in Germany.
109	Mt	Meitnerium	[278]	transition metal	Made in labs; not found in daily life	Named after physicist Lise Meitner.
110	Ds	Darmstadtium	[281]	transition metal	Made in labs; not found in daily life	Named after Darmstadt, the German city where it was made.
111	Rg	Roentgenium	[282]	transition metal	Made in labs; not found in daily life	Named after Wilhelm Roentgen, who discovered X-rays.
112	Cn	Copernicium	[285]	transition metal	Made in labs; not found in daily life	Named after astronomer Nicolaus Copernicus.
113	Nh	Nihonium	[286]	post-transition metal	Made in labs; not found in daily life	Nihon means Japan; it was the first element discovered in Asia.
114	Fl	Flerovium	[289]	post-transition metal	Made in labs; not found in daily life	Named after the Flerov laboratory in Russia.
115	Mc	Moscovium	[290]	post-transition metal	Made in labs; not found in daily life	Named after the Moscow region of Russia.
116	Lv	Livermorium	[293]	post-transition metal	Made in labs; not found in daily life	Named after the Lawrence Livermore laboratory in California.
117	Ts	Tennessine	[294]	halogen	Made in labs; not found in daily life	Named after the US state of Tennessee.
118	Og	Oganesson	[294]	noble gas	Made in labs; not found in daily life	Named after Yuri Oganessian, who was alive when it was named.

Next step

Hang the poster, then use one infographic a week to start class. Ask students to find the element in their classroom or at home.